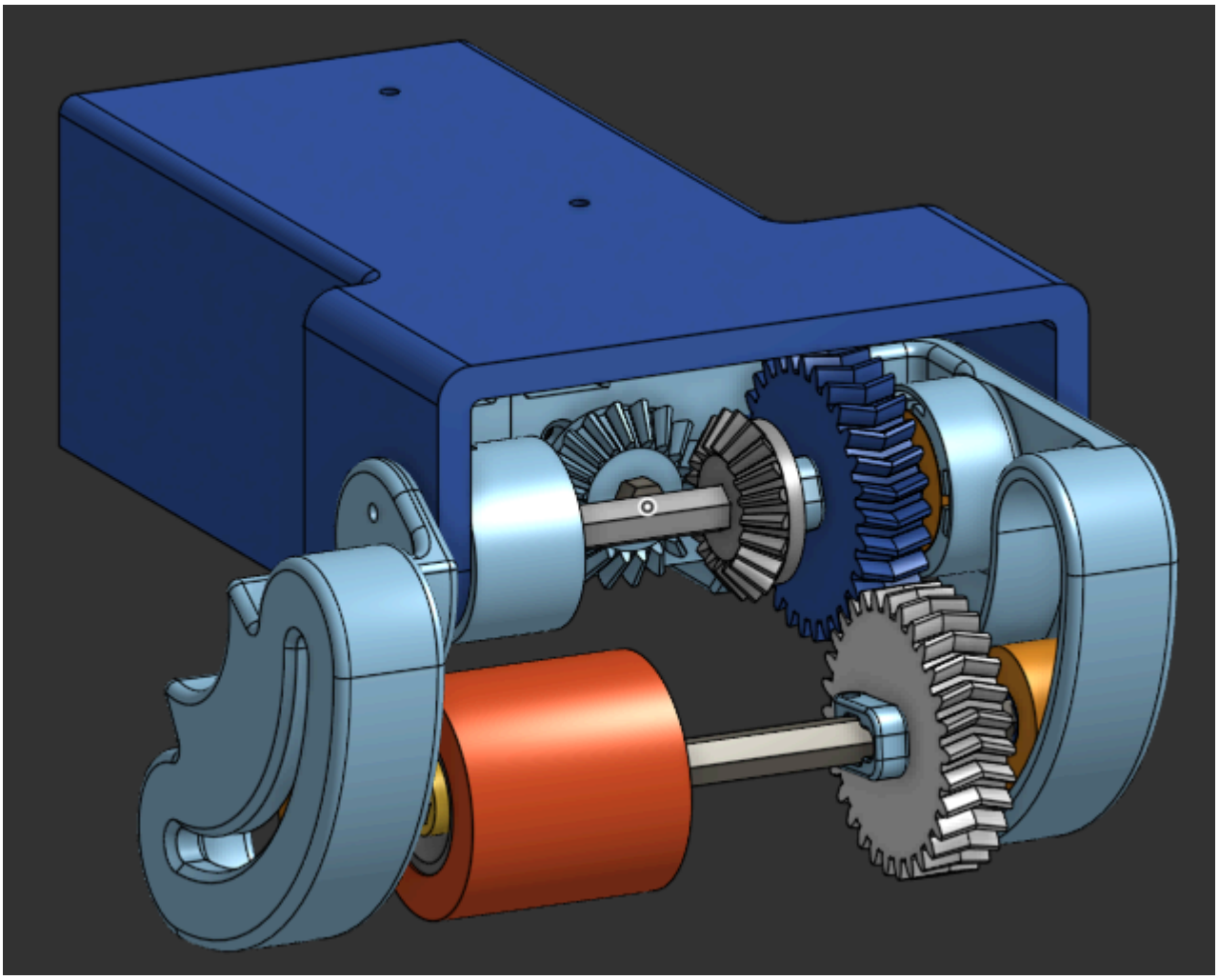
Design Report

Mahendra Kumar Gupta Potti



Interpretation

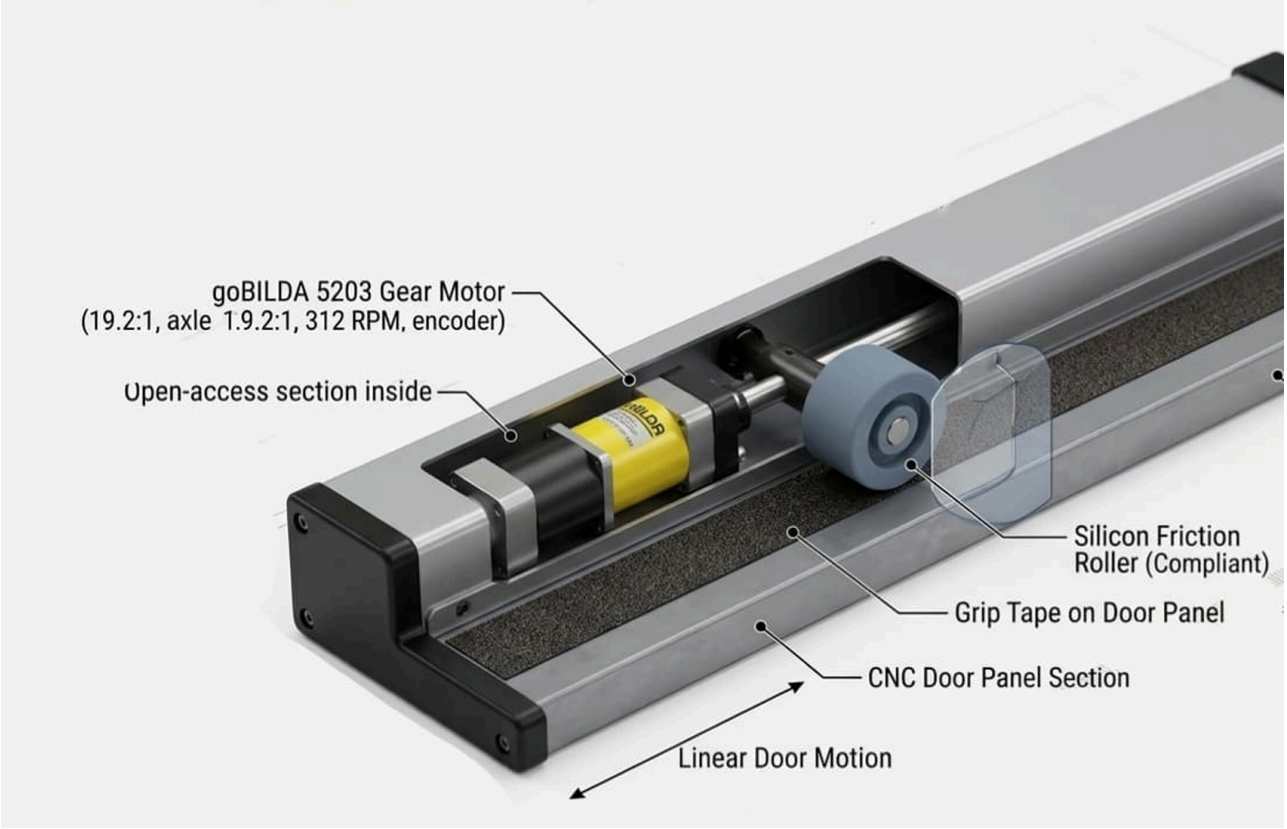
The problem statement asks to build a mechanism that allows us to automate the doors on a CNC machine. For simplicity, the scope of this assignment is to actuate a single door that moves strictly horizontally without pivoting like a van door.

Ideation

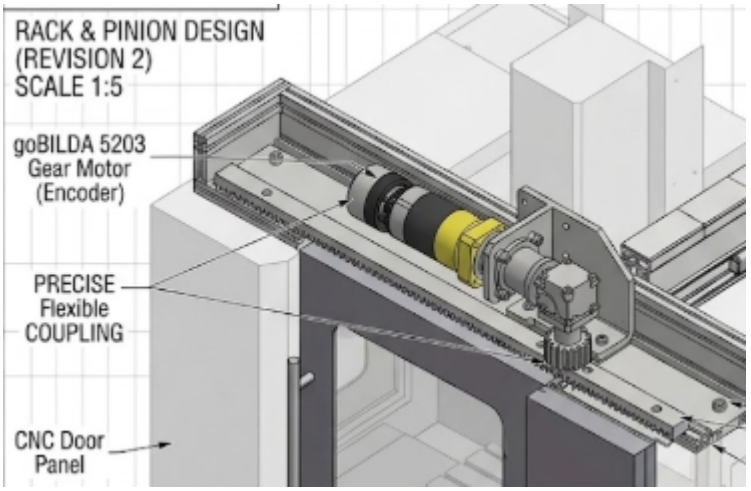
Concepts considered:

1. A soft **friction roller** (made of silicone or some other compliant material) that interacts with a grip tape pasted on the door panel. This would make it infinitely scalable as the mechanism is of a fixed

size and only the length of grip tape needs to be adjusted based on the length of the door.

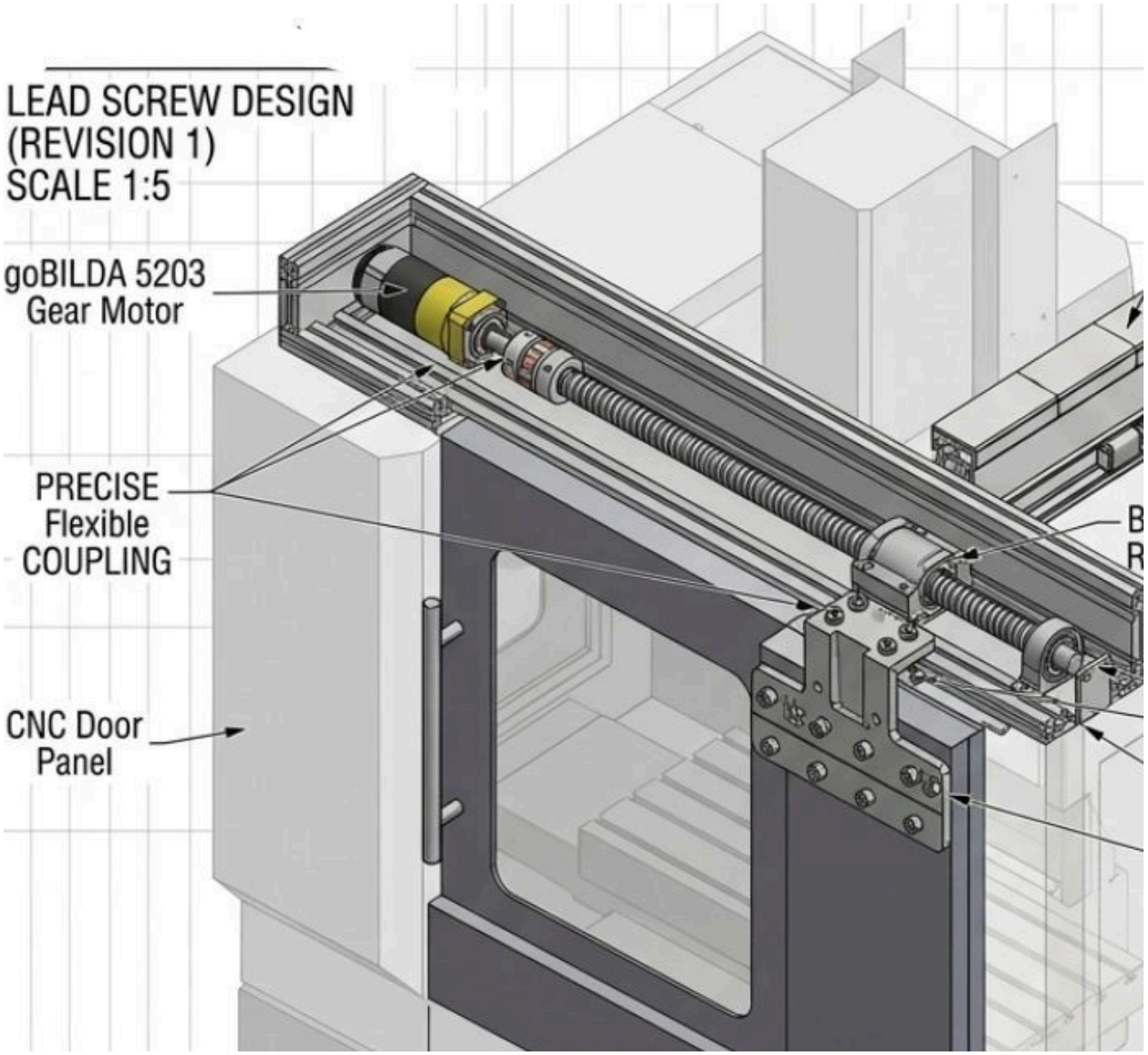


1. **Rack and pinion** mechanism where the rack is attached to the door and the motor drives the pinion. This limits scalability and the ability to actuate double doors. Rejected as very long racks are impractical

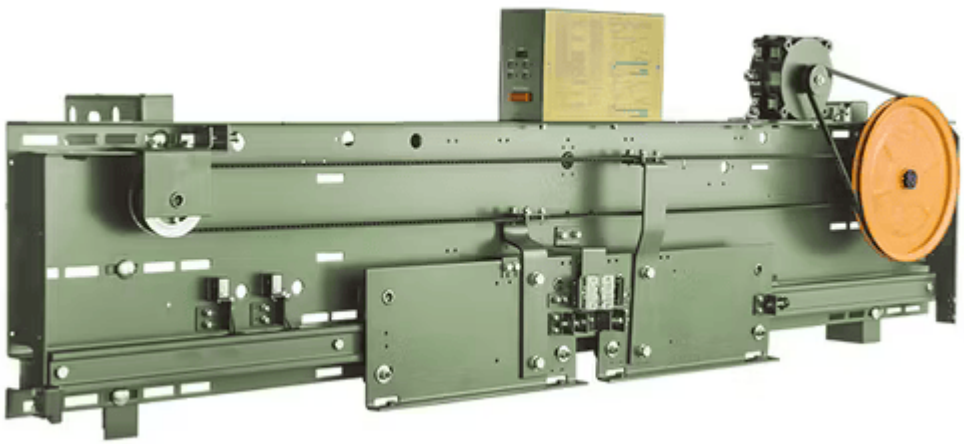


1. **Lead screw** mechanism. The motor is connected to a lead screw (or ball screw) that has a runner which is attached to the door either with a drop arm that attaches to the provided handle or using a lug fixture to the top of the door. Rejected as long lead screws can be costly and due to the overall

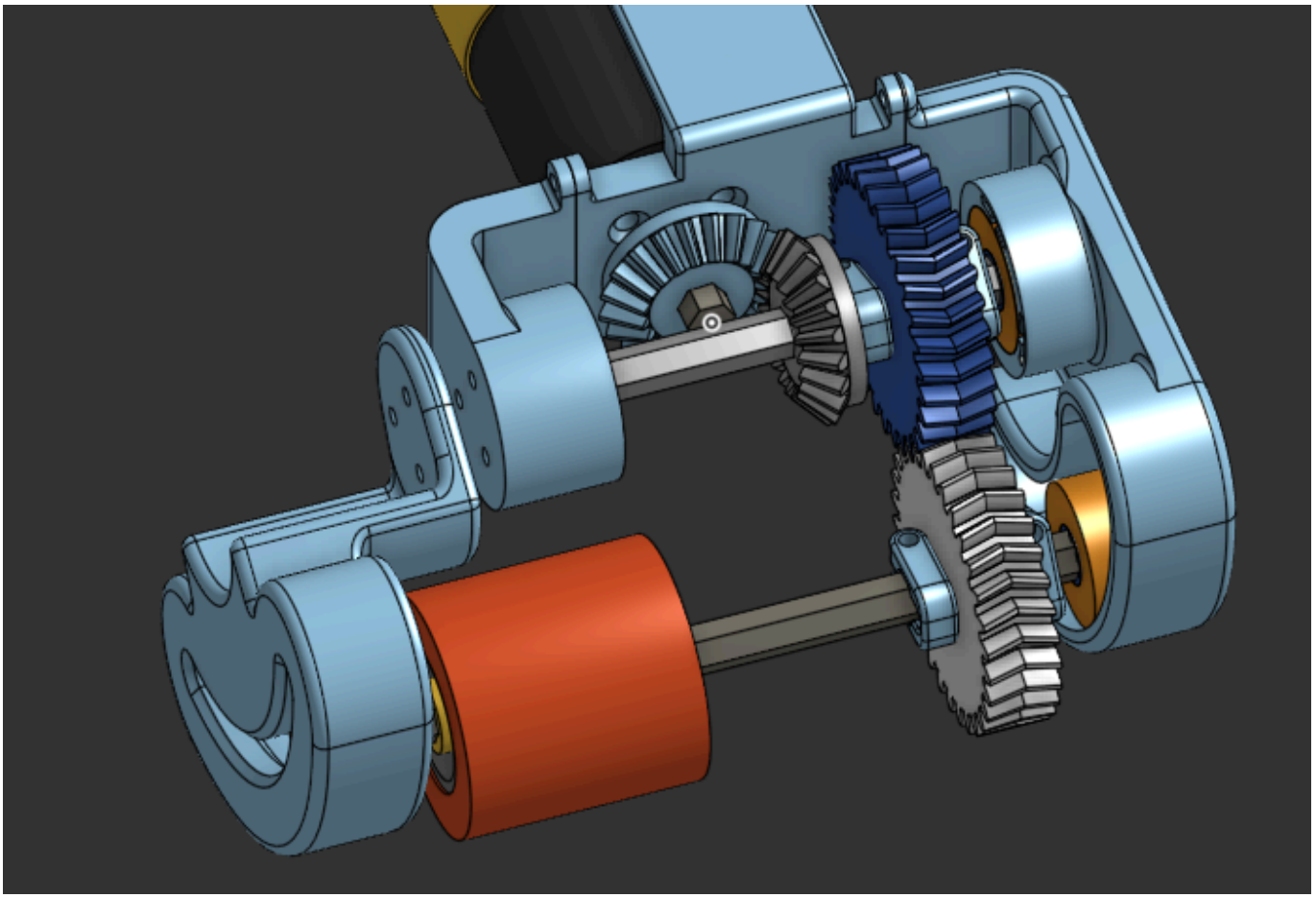
size of the mechanism.



1. **Belt drive** mechanism similar to the ones seen in automatic lift doors and in mall entrance doors
   Rejected due to the complexity



Selected concept: Friction Roller



I selected friction roller over the other well established choice like the belt mechanism because of simplicity and novelty. The roller is fixed in a 90° circular slot. The roller swings vertically by 45mm which ensures that the roller can contact the top edge of the door for different machines. The motor shaft is connected to a bevel system which allows the motor to be mounted in a compact orientation (along the door). There is a cutout in the roller's guide slot so that a bolt can be used to clamp the roller onto the door's top surface for grip. The bounding volume of the mechanism is 220mm x 160mm x 105mm.

Engineering Assumptions

1. The door rolls on a fixed bottom rail with linear roller bearings.
2. The environment is reasonably non ideal: airborne dust, metal chippings, lubricants, coolants and human sweat are considered.
3. The door panel has a flat top surface and is accessible across the full length.
4. The door panel is sturdy enough to allow some downward force on the top surface (for clamping the roller)
5. The top of the machine is flat and allows for the mechanism to be fixed (via bolting or clamping)

Calculations

Assumptions:

Parameter	Value	Note
Stroke length	1600 mm	Worst case value

Parameter	Value	Note
Target open/close time	5s	320 mm/s avg
Motion profile	0.8s accel 3s const. vel. 1.2s decel	
Existing door rail friction coeff.	0.003 to 0.005	conservative estimate for a well lubricated rail source
Roller to tape friction coeff.	1.05 to 1.35	Given by 3M for their anti slip tapes
Gear transmission efficiency	85%	conservative estimate
Motor parameters	Speed: 312 RPM Max torque: 24.3 kgcm	

calculations

Scalability

The design is very scalable as the roller can roll on the top of the door regardless of its length. If the available mounting surface is more than 45mm away from the top of the door then the given roller slot will have to be changed to fit the required distance.

Safety and Maintenance

There is an aspect of maintenance as the grip tape and roller will wear down with usage. But the mechanism can be compared to a skateboard where even a weathered wheel will grip the road surface, so the grip tape would need replacement periodically. Additionally, with usage, the roller slot clamping might reduce (due to hertzian stess if the door is not very flat) and would require tightening periodically.

AI Usage

Generative AI was only used in the following places:

1. Assist in ideation and in picking constants (like sizing, friction etc)
2. Generate concept sketches
3. Proofread documents

The usage is limited to assisting the actual design process, not as the main designer. The documents are also hand-written, using Markdown format and exported as PDFs